

Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 5: Volumes of Revolution

Chapter 5: Volumes of Revolution

Why this matters

This chapter takes the integration skills from A Level Maths and applies them to a genuinely new, visual problem: finding the volume of a solid formed by spinning a curve around an axis. It's a short chapter, but it rewards you for setting up the integral carefully — most marks are lost not in the calculus itself, but in getting the formula and limits right at the start.

Don't confuse this with the A Level area-under-a-curve integral $\int y \, dx$ — that gives a flat 2D area with no π and no squaring; this chapter's $V = \pi \int y^2 \, dx$ gives a 3D volume. Mixing the two up (e.g. forgetting to square y , or adding a stray π to an area question) is a very common slip when the two topics are studied close together.

Concept: volume of revolution about the x-axis

When the region between a curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ is rotated fully (360°) about the x-axis, it sweeps out a solid of revolution. Its volume is given by:

$$V = \pi \int_a^b y^2 \, dx$$

This formula comes from approximating the solid as a stack of thin circular discs, each of radius y and thickness dx , with volume $\pi y^2 \, dx$, then summing (integrating) over all of them. You are assumed to already be confident integrating polynomials, trig functions, exponentials and using substitution from A Level Maths — the new content here is purely the setup and application of this volume formula.

This formula only works when the region being rotated lies entirely on one side of the axis — it never crosses it. If a curve dips below the x-axis and back above it within $[a, b]$, the y^2 term makes the negative part positive again, silently adding volume from a region that shouldn't be there. Always sketch the curve first and check it doesn't cross the axis of rotation; if it does, split the integral at the crossing point(s) and treat each piece separately.

Key vocabulary

Term	Meaning
Solid of revolution	The 3D solid formed when a 2D region is rotated fully about an axis
Volume about the x-axis	$V = \pi \int_a^b y^2 \, dx$
Volume about the y-axis	$V = \pi \int_c^d x^2 \, dy$
Limits of integration	The x (or y) values between which the region is bounded

Worked example 1: revolution about the x-axis
EXAMPLE 1

Question. The region bounded by the curve $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 2$ is rotated 360° about the x-axis. Find the volume of the solid formed, giving your answer in terms of π .

Step 1. Write down the volume formula and substitute $y = x^2$: $V = \pi \int_0^2 (x^2)^2 dx = \pi \int_0^2 x^4 dx$. **(M1)**

Step 2. Integrate: $\int x^4 dx = x^5/5$. **(M1)**

Step 3. Apply the limits: $\pi[x^5/5]_0^2 = \pi(2^5/5 - 0) = 32\pi/5$. **(A1)**

Answer: $V = 32\pi/5$ cubic units.

Exam tip
EXAM TIP

The most common setup error is forgetting to square y before integrating — integrating y itself rather than y^2 . Get into the habit of writing " $V = \pi \int y^2 dx$ " as a fixed opening line for every question, then substituting, so the squaring step is never skipped.

Concept: volume of revolution about the y-axis

By symmetry, rotating a region about the y-axis instead uses discs perpendicular to the y-axis, giving $V = \pi \int c^d x^2 dy$. This requires the curve's equation to be rearranged to make x (or x^2) the subject in terms of y , and the limits become y -values, not x -values.

Worked example 2: revolution about the y-axis
EXAMPLE 2

Question. The region bounded by the curve $y = \sqrt{x}$, the y-axis, and the lines $y = 1$ and $y = 3$ is rotated 360° about the y-axis. Find the volume of the solid formed.

Step 1. Rearrange $y = \sqrt{x}$ to make x the subject: $x = y^2$, so $x^2 = y^4$. **(M1)**

Step 2. Set up the integral: $V = \pi \int_1^3 y^4 dy$. **(M1)**

Step 3. Integrate: $\int y^4 dy = y^5/5$, so $V = \pi[y^5/5]_1^3$. **(M1)**

Step 4. Apply the limits: $\pi(3^5/5 - 1^5/5) = \pi(243/5 - 1/5) = \pi(242/5)$. **(A1)**

Answer: $V = 242\pi/5$ cubic units.

Exam tip
EXAM TIP

It is common to rearrange correctly for x but then forget to square it, integrating x instead of x^2 . It helps to write the rearranged equation for x^2 explicitly as its own line (as in Step 1 here) before

ever writing the integral, rather than trying to square inside the integral sign where errors hide more easily.

Concept: volumes involving subtraction (regions between curves or hollow solids)

Some questions ask for the volume between two curves rotated about an axis — for example, the region enclosed between $y = f(x)$ and $y = g(x)$. Rotate each curve separately and **subtract the smaller resulting volume from the larger**:

$$V = \pi \int (y_1^2 - y_2^2) dx$$

where y_1 is the curve further from the axis of rotation.

Worked example 3: volume between two curves

EXAMPLE 3

Question. The region enclosed between the curves $y = x$ and $y = x^2$ for $0 \leq x \leq 1$ is rotated 360° about the x -axis. Find the volume of the solid formed.

Step 1. Identify which curve is further from the x -axis on $[0, 1]$: since $x \geq x^2$ on this interval, $y = x$ is the outer curve. **(B1)**

Step 2. Set up the subtraction integral: $V = \pi \int_0^1 (x^2 - (x^2)^2) dx = \pi \int_0^1 (x^2 - x^4) dx$. **(M1 A1)**

Step 3. Integrate: $\int (x^2 - x^4) dx = x^3/3 - x^5/5$. **(M1)**

Step 4. Apply the limits: $\pi[(1/3 - 1/5) - (0 - 0)] = \pi(5/15 - 3/15) = \pi(2/15)$. **(A1)**

Answer: $V = 2\pi/15$ cubic units.

Exam tip

EXAM TIP

Remember that you must square each curve BEFORE subtracting — the mistake of writing $(y_1 - y_2)^2$ instead of $y_1^2 - y_2^2$ is extremely common and completely changes the answer. A quick sketch showing the two "washers" (annulus cross-sections) makes clear why it must be the difference of the two squared radii.

Chapter summary

Idea	Key point
Rotation about the x -axis	$V = \pi \int_a^b y^2 dx$
Rotation about the y -axis	$V = \pi \int^c d x^2 dy$; rearrange the curve to make x the subject first
Between two curves	$V = \pi \int (y_1^2 - y_2^2) dx$, outer curve squared minus inner curve squared

Setup is the key skill

The calculus is often familiar; the volume formula and limits are the new part

Common exam mistakes

- Forgetting to square y (or x) before integrating.
- Forgetting the factor of π in the final answer.
- Using the wrong limits (x -limits when rotating about the y -axis, or vice versa).
- When subtracting two curves, squaring the difference $(y_1 - y_2)^2$ instead of taking the difference of the squares $y_1^2 - y_2^2$.
- Sign errors when substituting negative limits, or forgetting the constant of integration is not needed for definite integrals.

You should now be able to

- Derive and apply the formula for the volume of revolution about the x -axis.
- Rearrange a curve's equation to find the volume of revolution about the y -axis.
- Set up and evaluate the volume of a solid formed by rotating the region between two curves.
- Sketch the relevant region and solid to check that a volume calculation is set up correctly.